Water Turbidity Results– Outline

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August 2026

This report provides data on in silica measurements of photon scattering in turbid water. It utilizes Monte Carlo Extreme[] (MCX) to simulate photons in medium with varying turbidity levels. This report serves to fill the gap in literature between instrumentation design, optimization, and standardization of optical turbidity sensors.

1 Background

Turbidity is a measure of the haziness of a fluid due to particulates within it.It serves as a strong indicator of suspended solids in water. Nephelometric Turbidity Unit, NTU, is the standard measurement unit for turbidity and represents the how much light scatters 90 degrees from the beamline. There is no direct conversion between turbidity and scattering coefficient because turbidity measures the 90 degree reflection only.

Measurements of turbidity are important due to a host of reasons given in [][][]. One of these resources discusses the reduction of shallow water vegetation due to the highly turbid waters in Chesapeake bay. It is important to keep track of water turbidity to understand the change in water sediment levels, phytoplankton levels, and take preemptive action to prevent long lasting harm.

1.1 Instrumentation background

The Secchi Disk is a 30 cm black and white disk. A string is attached to the disk and is lowered into the water. When the visibility of the water does not allow for distinguishing between the black and the white sections of the disk, the depth is recorded. This depth is related to the turbidity. This method is widely used for measuring the turbidity of open waters.

Most modern turbidity meters have a light source and a detector photodiode positioned 90 degrees from the light source. Some meters also have a detector at 180 in order to measure the transmittance loss alongside the scattering. Constructing a turbidity meter and obtaining accurate measurements involves optimizing for numerous parameters such as the z-distance of the of beam detector, x-distance of the off beam detector, Op-Amp gain, source beam profile, detector acceptance angle, and external noise. While these may be solvable, the problem gains three extra dimensions when considering the variability

of absorption coefficient, scattering coefficient, and volume scattering function of the medium, namely the optical properties of water (IOPs). The design becomes a "Chicken and Egg" problem as it must be designed to measure a range of values that it is being constructed to measure. While constructing a detector able to function throughout a wide range IOPs serves as a simple solution, the cost and complexity of the design lands itself out of the intended clientele's budget in nature conservation.

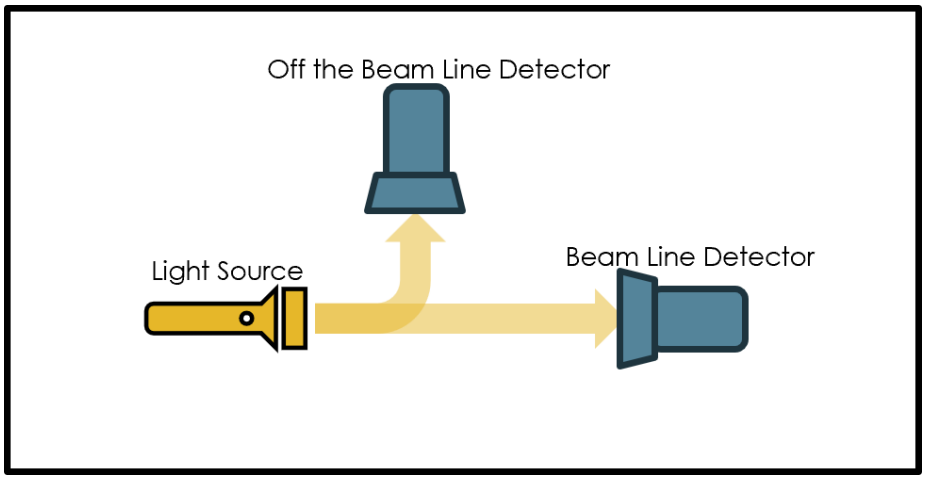


Figure 1: This is the geometry of a simple turbidity meter. Detectors along and off the beam line measure the scattered and transmitted light.

2 Theory

2.1 Inherent Optical Properties

Inherent Optical Properties of matter are the absorption coefficient (μ_a), scattering coefficient (μ_s), and volume scattering function ($\beta(\psi)$). These describe the interaction of matter with radiation. Absorption coefficient describes the number of absorption events that occur per unit distance traveled. Scattering coefficient describes the number of scattering events that occurs per unit distance traveled. We use $\frac{\text{event}}{mm}$ for these coefficients. The volume scattering function describes the distribution of scattering angles (ψ) with respect to the incident beam (ξ'). Below are representations of a single absorption and scattering events.

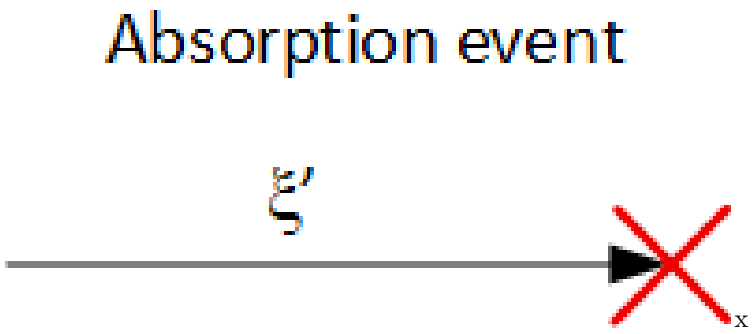


Figure 2: Absorption Event Conceptualization for a single photon

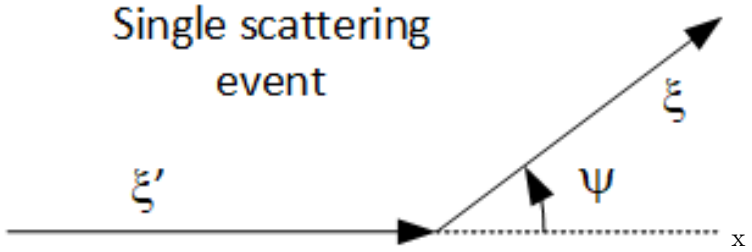


Figure 3: Scattering Event Conceptualization

2.2 Volume Scattering Functions

Volume Scattering Functions (VSF) describe the probability distribution of the scattering direction upon an individual scattering event. An isotropic VSF carries equal probability to scatter in any direction upon a scattering event. If a scattering event barely changes the initial direction of travel, it is anisotropic.

In this report we consider two volume scattering functions, Henry-Greenstein (HG) Volume Scattering Function and Fornier-Forand (FF) Volume Scattering Function, and we decide to proceed with Fornier-Forand Volume Scattering function.

Henry-Greenstein function derives from Mie theory and assumes uniform distribution of uniformly sized particles. Henry-Greenstein is written as a function of g, the anisotropy factor. Anisotropy is a number between 0 and 1.It is the average of the cosine of the distribution function. As such, a high g factor implies that the scattering is forvard biased. HG is widely used in industry and Monte Carlo methods because it has a closed form analytic inverse, which is needed for efficient sampling of the distribution.

$$p_{\mathrm{HG}}(\psi) = \frac{1-g^2}{4\pi\left(1+g^2-2g\cos\psi\right)^{3/2}} \tag{1}$$

$$g = \langle\cos\psi\rangle = 2\pi\int_0^\pi p(\psi)\cos\psi\sin\psi\,d\psi \tag{2}$$

The Fornier-Forand modifies HG by considering the index of refraction of the suspended particles as well as the size distribution. This function very closely approximates Petzold's measurements[] of Volume Scattering Function in natural waters. FF approximates the measurements significantly closer near small degrees of scattering. For processes in turbid water where most scattering events divert minimally from their direction of travel($\psi < 2°$), these small angles became critical. This distribution is given by $\beta_{FF}(\psi)$.

$$\nu = \frac{3-\mu}{2} \tag{3}$$

$$\delta(\psi) = \frac{4}{3(n-1)^2}\sin^2\left(\frac{\psi}{2}\right) \tag{4}$$

$$\begin{aligned}\beta_{FF}(\psi) = \frac{1}{4\pi(1-\delta)^2\delta^\nu}\Bigg[&\nu(1-\delta)-(1-\delta^\nu)\\ &+\left(\delta\left(1-\delta^\nu\right)-\nu(1-\delta)\right)\sin^{-2}\left(\frac{\psi}{2}\right)\Bigg]\\ &+\frac{1-\delta_\pi^\nu}{16\pi(\delta_\pi-1)\delta_\pi^\nu}\left(3\cos^2\psi-1\right).\end{aligned} \tag{5}$$

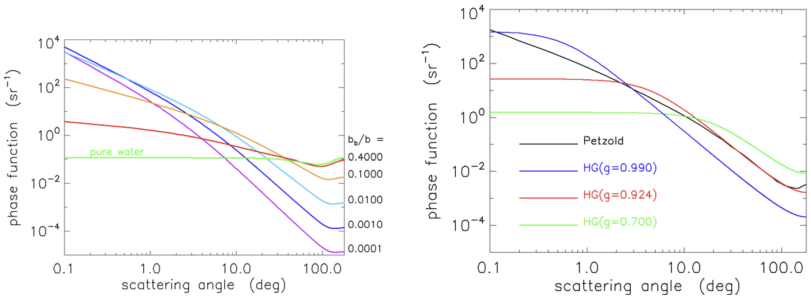


Figure 4: Comparison of Fornier Forand and Henry Greenstein Volume Scattering Functions (VSF)

As a two parameter function, FF has multiple combinations of n and junge that yield the same function, as such, back-scatter fraction is commonly used

as the defining parameter. The same parameter is also often used to define HG. [Ocean Optics]

$$B_{\mathrm{HG}} = \frac{1-g}{2g}\left(\frac{1+g}{\sqrt{1+g^2}} - 1\right) \tag{6}$$

$$B_{\mathrm{FF}} = 1 - \frac{1 - \delta\left(\frac{\pi}{2}\right)^{v+1} - \frac{1}{2}\left(1 - \delta\left(\frac{\pi}{2}\right)^{v}\right)}{\left(1 - \delta\left(\frac{\pi}{2}\right)\right)\delta\left(\frac{\pi}{2}\right)^{v}} \tag{7}$$

The figure below displays a comparison of the two functions with the same back-scattering fraction of 0.0144. This value is used as it is near commonly reported back scatter values for turbid water.

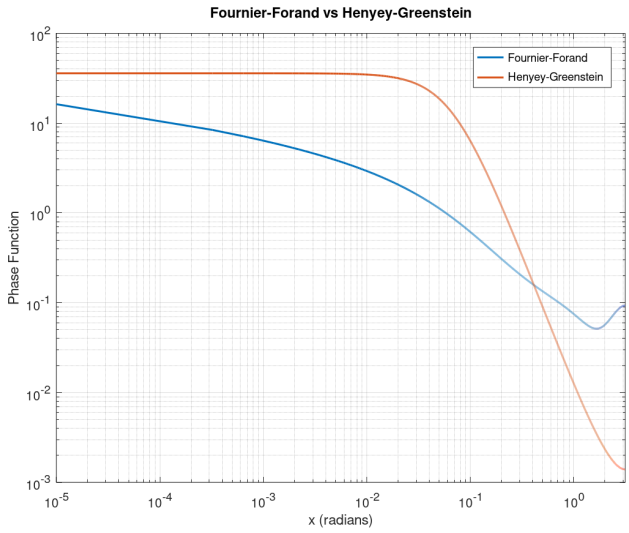


Figure 5: Comparison of Fornier Forand and Henry Greenstein VSF at equal back scatter $B_{HG} = B_{FF} = 0.0144$

2.3 High Optical Depth and Monte Carlo

Given the scattering coefficient, absorption coefficient, and the parameters defining the Fornier-Forand, distribution, a Monte Carlo distribution allows us to solve for the intensity of light at various angles and distances from a light source. At high optical depth, which refers to the regime where multiple scattering and absorption events occur, it is important to note that the Monte Carlo results will not approximate the Fornier-Forand distribution. This is because the VSF only describes the distribution for a single scattering event. The accumulation of these events lead to a different distribution at various distances.

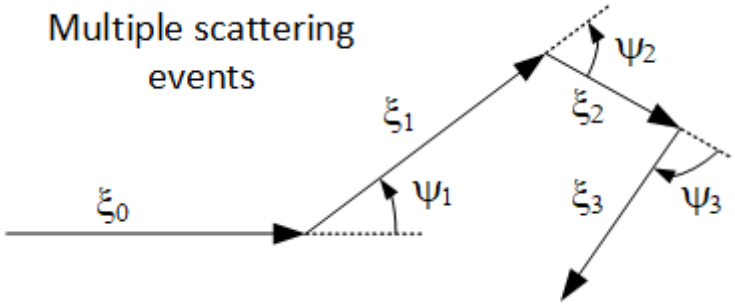


Figure 6: Multiple Scattering Region where an analytical distribution over angles and distances become computationally unreasonable and a MonteCarlo simulation is needed.

3 Computation

This report utilizes Monte Carlo Extreme (MCX) software to run Monte Carlo Simulations on eight Tesla K80 GPGPUs. The MCX workflow is as follows[3]

1. A photon is first launched at the position of the source along an incident direction vector with an initial packet weight of 1.
2. A scattering length, the distance to the next scattering event site, is computed using the scattering coefficient of the current voxel based on an exponential distribution.
3. The photon packet is stepped one voxel length along the scattering trajectory (if the remaining scattering length is less than one voxel, stop at the end of the trajectory).
4. The packet weight is reduced by the absorption coefficient along that step.
5. The packet weight is added to its current voxel's raw probability at a fixed time interval and temporally binned based on user specified time gates.
6. Repeat from step 3 until the photon has traveled the total scattering length.
7. Calculate a new scattering direction vector: the azimuth and zenith scattering angles are computed, respectively, by a random number uniformly distributed in$[0, 2\pi)$ and a random number between $[0, \pi]$ following the determined phase function.
8. Repeat from step 2 until the photon exits the domain or reaches the maximum time-gate.
9. Repeat from step 1 until the total number of packet steps or simulated photons is reached.

The simulation parameters used are given in the tables below for the two setups. The first setup is the large volume setup where the source is placed at the center of the volume and the analysis is preformed around a circle around the source at a plane slice. The second setup is constructed in direct reference to figure 1, the turbidity meter. The source is placed upon the -x face and is pointed towards the +x face. The domain is a rectangle and the photons leaving the domain through a 1 mm strip going through the centers of the +y and +x faces are recorded and binned based on position.

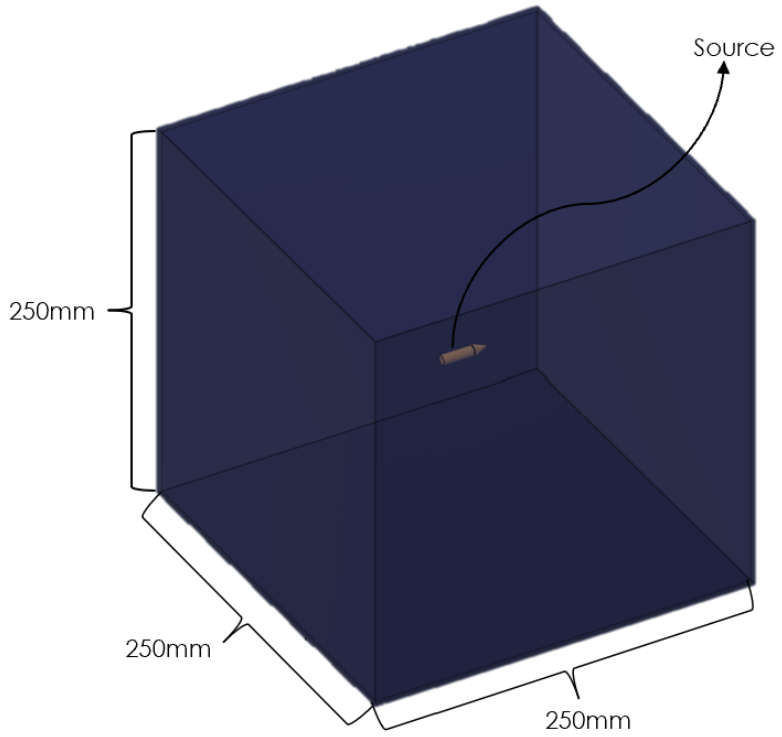


Figure 7: This is the geometry of Setup 1 which is a $250cm^3$ homogeneous domain with a central pencil beam source

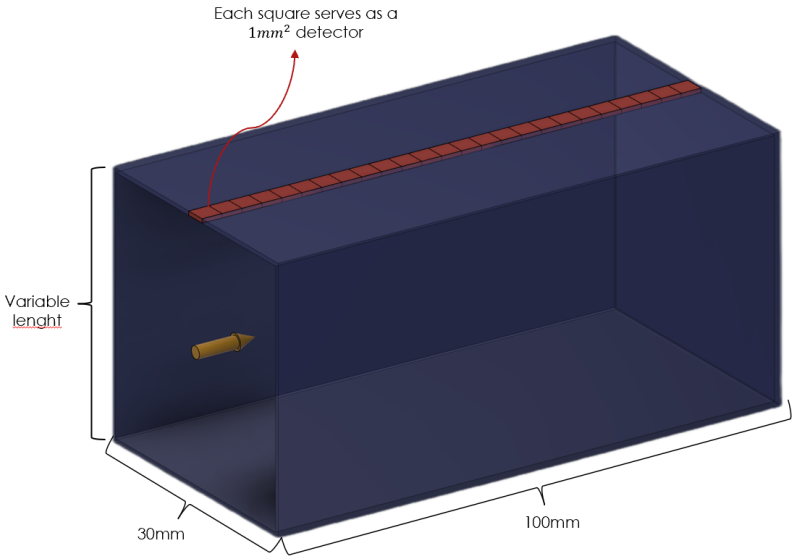


Figure 8: This is the geometry of the setup 2, which has detectors facing the beam line from an orthogonal direction, shown by the red squares, at variable distance

Table 1: Monte Carlo Parameters - Setup 1

Parameter	Value
Number of photons	10^7
Domain size	$500 \times 500 \times 500$ voxels
Voxel size	1 mm
Source position	$(250, 250, 250)$
Source direction	$(0, 0, 1)$
Absorption coefficient, μ_a	specified on runtime
Scattering coefficient, μ_s	specified on runtime
Phase function	Fournier–Forand
Junge parameter, μ	3.56
FF relative refractive index, n_{FF}	1.09
Number of photons	10^7
Boundary condition	Absorbing
Time gate	0–5 ns

Table 2: Monte Carlo Parameters - Setup 2.

Parameter	Value
Number of photons	10^7
Domain size	$100 \times 30 \times ...$ voxels
Voxel size	1 mm
Source position	$(1, 15, 50)$
Source direction	$(1, 0, 0)$
Absorption coefficient, μ_a	$0.001, 0.0015, ..., 0.005$
Scattering coefficient, μ_s	$0.01, 0.02, ..., 0.1$
Phase function	Fournier–Forand
Junge parameter, μ	3.56
FF relative refractive index, n_{FF}	1.09
Number of photons	10^7
Boundary condition	Absorbing
Time gate	0–5 ns

4 Results

The results section shares findings from the experimental setup 2 given in figure 8.

4.1 Optimal position of the detector along the x axis given the detector height, μ_a, μ_s, and $\beta(\psi)$

This example assumes $\mu_s = 0.04$, $\mu_a = 0.003$, $\beta(\psi)$ as the Fornier Forand function with $B_{FF} = 0.0144$ and $z = 70mm$ which yields an off beam distance of $20mm$ given the source is positioned at $z = 50mm$. In the figure below the x-axis represents the position where photons exit the detector surface which spans the entire axis. The x axis values here are in direct reference to the red squares shown in figure 1. The y-axis, labeled Escaping Photon Weight, is the sum of the remaining statistical weights of all photons exiting within each spatial bin. Escaping photon weight can be interpreted as a relative detected intensity. This graph shows that for the given IPOs and detector height, best placement for a detector, optimizing for maximum radiance, is $100mm$ away from the source in the x axis, where the peak appears.

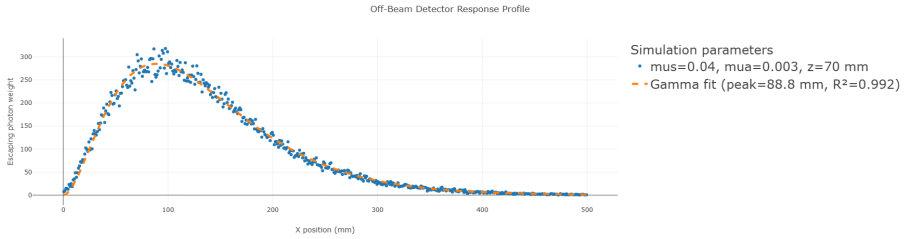


Figure 9: For given IPOs and z position of the off the beam detector ($z = 70mm$), the relative intensity of light at different x positions where the detector could be placed. This one suggests 88.8mm along the x axis maximizes the relative intensity.

This naturally raises a secondary question of weather this $100mm$ distance is static with respect to the change in the detector height, z. The following 2 figures , 10 and 11, show that this number is dependent on the z coordinate.

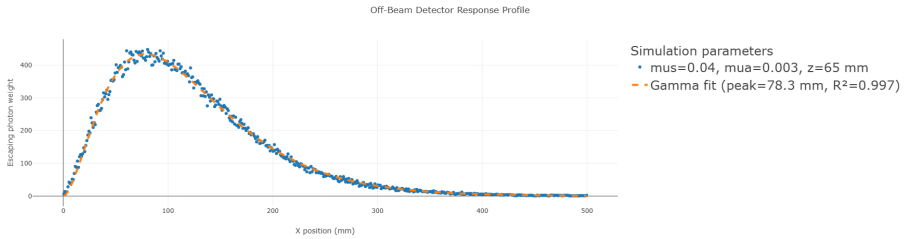


Figure 10: For given IPOs and z position of the off the beam detector ($z = 65mm$), the relative intensity of light at different x positions where the detector could be placed. This one suggests 78.3mm along the x axis maximizes the relative intensity.

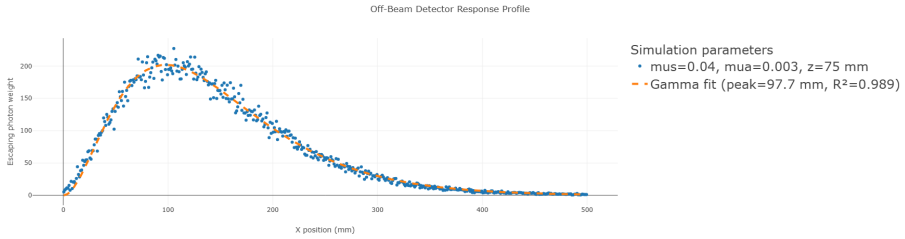


Figure 11: For given IPOs and z position of the off the beam detector ($z = 75mm$), the relative intensity of light at different x positions where the detector could be placed. This one suggests 97.7mm along the x axis maximizes the relative intensity.

Figure 12 below shows the optimal detector position on the x axis which yields the highest Escaping Photon Weight. It estimates this position by fitting the detector response profile to a gamma distribution, $f(x; k, \theta) = \frac{1}{\Gamma(k)\,\theta^k}$, and calculating its peak.

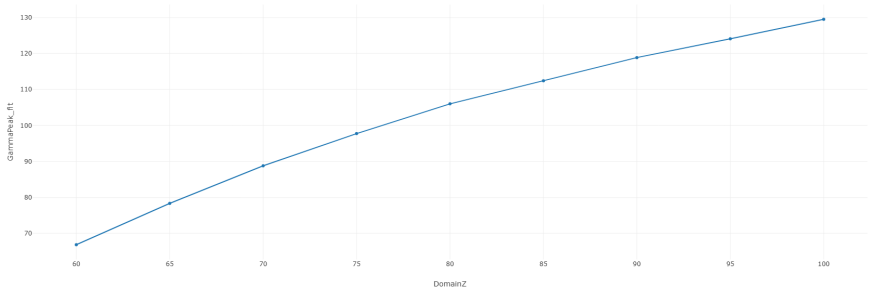


Figure 12: Optimal detector position along the x axis given the z height

The following graph provides the same information for different μ_a values.

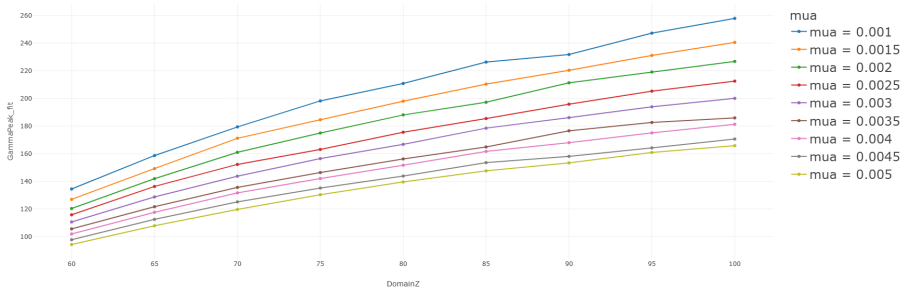


Figure 13: Optimal detector position along the x axis as a function of detector height z for $\mu_a = 0.001$–0.005 mm^{-1} ($\mu_s = 0.04$ mm^{-1}, $B_{FF} = 0.0144$).

It is important to notice that while this graph shows the ideal location for a given z while optimizing for the radiance, it does not yield any information regarding the height of that peak. The following figure shows multiple of the distribution graphs overlapping.

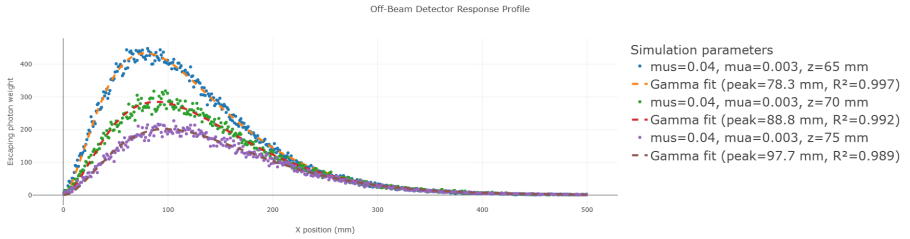


Figure 14: Off-beam detector response profiles at detector heights $z = 65$, 70, and 75 mm ($\mu_s = 0.04$ mm^{-1}, $\mu_a = 0.003$ mm^{-1}). As z increases, the peak intensity drops and shifts to larger x.

The height of this maximum peak given the IPOs and z is shown by the following graph in figure 15. It is important to re-iterate that each of the points in this graph tells the maximum relative intensity achievable given the person chooses the optimal position for the detector on the x axis. As theory would suggest, the graphs resemble an exponential decay.

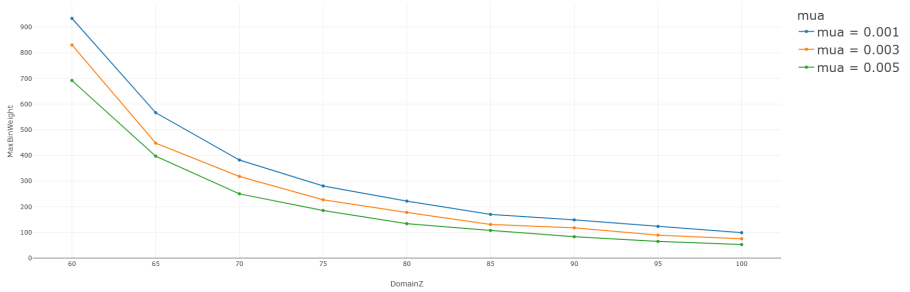


Figure 15: Maximum escaping photon weight (peak of the gamma fit at the optimal x position) as a function of off-beam detector height z, for $\mu_a = 0.001$, 0.003, and 0.005 mm^{-1} ($\mu_s = 0.04$ mm^{-1}, $B_{FF} = 0.0144$). The achievable signal decays roughly exponentially as the detector moves away from the beam line.

Below is the output of a simulation ran with unrealistically high μ_s to provide a convenient visual to understand the analysis discussed above. In this image, the black vertical line proposes a detector $5mm$ above the beamline. In this case, the highest intensity point on the x axis is shown with a looking glass. On the other hand, the red line proposes a detector $2.5mm$ above the beamline. In this case the highest intensity point is marked with a red looking glass. It must be noted that despite the lack of strong differentiation with color grading, moving the red looking glass away from the source would reduce the intensity due to beers law.

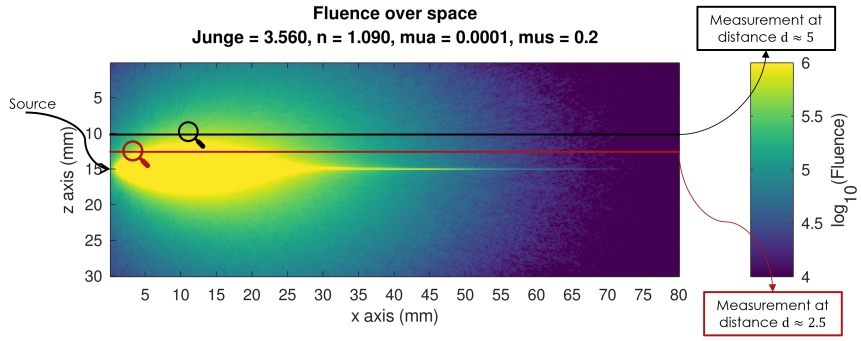


Figure 16: Illustrative fluence map with an unrealistically high scattering coefficient ($\mu_s = 0.2$ mm^{-1}, $\mu_a = 0.0001$ mm^{-1}). The black and red lines represent detector planes 5 mm and 2.5 mm above the beam line, and the magnifiers mark the position of maximum intensity along each.

4.2 Detector Sensitivity to Change in Turbidity

While a substantial design challenge for the detector orthogonal to the beam line is the amount of light it receives, another consideration would be maximizing its sensitivity to changes in turbidity or to the scattering coefficient of the water in a more simplified model. The results in this section follow the assumptions below:

- Turbidity is directly related to μ_s
- Turbidity has no relation to μ_a
- The Volume Scattering Function remains unchanged
- The values $0.02 \geq \mu_s \geq 0.1$ encompass reasonable variation in the observed system
- The rest of the simplifying assumptions made throughout this paper hold.

The figure below compares the detector distribution graphs of a system with $\mu_a = 0.003$,$B_{FF} = 0.0144$, and $z = 65mm$ for a range of different μ_s.

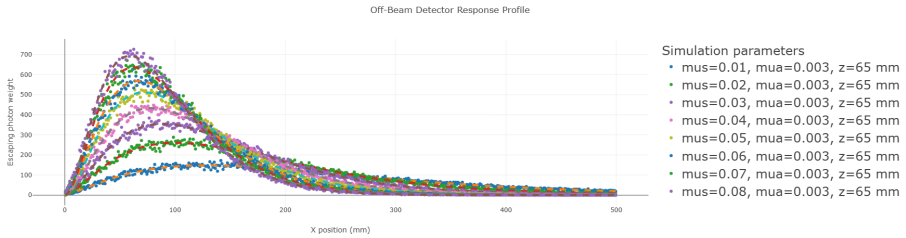


Figure 17: Off-Beam Detector Response Profiles for Different μ_s Values

The sensitivity $S(x, \mu_s)$ of the system at position x as the scattering coefficient goes from $a \to b$ is given by the the difference of the Distribution Functions $D(x, \mu_s = a)$ and $D(x, \mu_s = b)$.

$$S(x, \mu_s = a \to \mu_s = b) = D(x, \mu_s = a) - D(x, \mu_s = b)$$

4.2.1 Example Sensitivity Measurement

This example explores the sensitivity of the off the beam line detector with respect to change in μ_s. We fix other variables as shown in figure 17: $\mu_a = 0.003$, $z = 65mm$, source z position $= 50mm$,$B_{FF} = 0.0144$. Then, we select an x-position to place the detector. This example chooses $x = 78.3$ in order to maximize the escaped photon weight when the $\mu_s = 0.04$ as shwon in figure 18

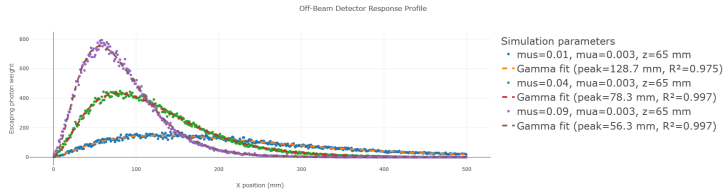


Figure 18: Clone of figure 17 with less values of μ_s. The green circle shows where the distribution for μ_s is at a maximum

The figure below serves as a reminder of the setup.

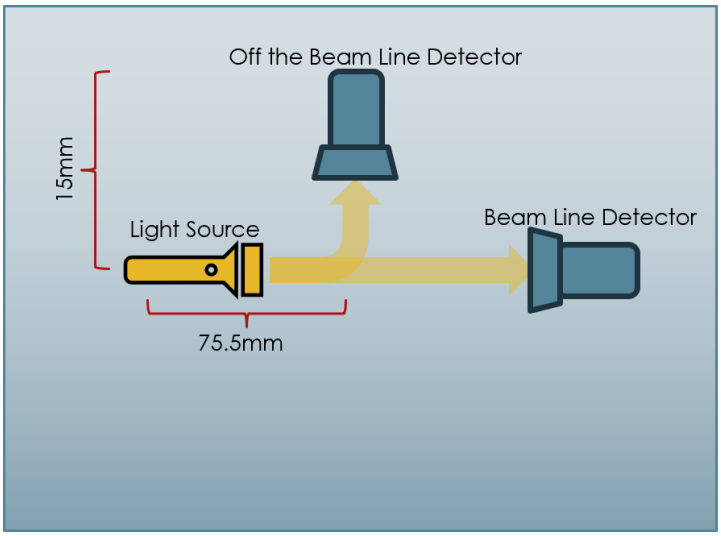


Figure 19: Setup 2, simplified 2 Dimensional view with the light source, detectors, and a representation of the scattered and transmitted light.

In order to find the sensitivity of the detector in this setup moving from $\mu_s = 0.04 \rightarrow 0.03$ we preform the following operation. Figure 20 shows this operation graphically.

$$S(x = 78.3, \mu_s = 0.04 \rightarrow 0.03) = D(x = 78.3, \mu_s = 0.04) - D(x = 78.3, \mu_s = 0.03) = 443 - 343 = 100$$

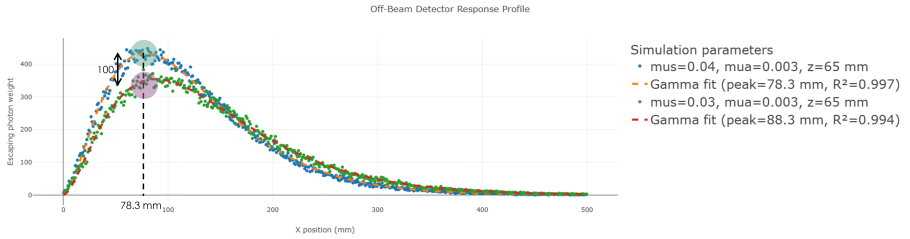


Figure 20: Sensitivity at the modeled position is the difference between the values in green and purple.

Measuring the escaped photon weight at along the same x value for other μ_s values and plotting it we reveal figure 21. Which shows a sublinear trend.

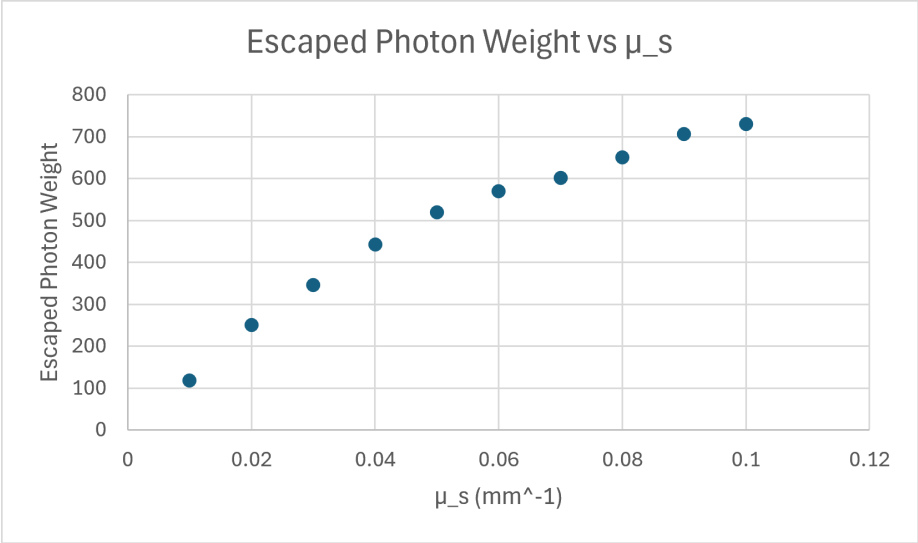


Figure 21: Enter Caption

Computing the difference between these values to uncover the sensitivity reveals figure 22. The shape of this graph does yield more to be explored.

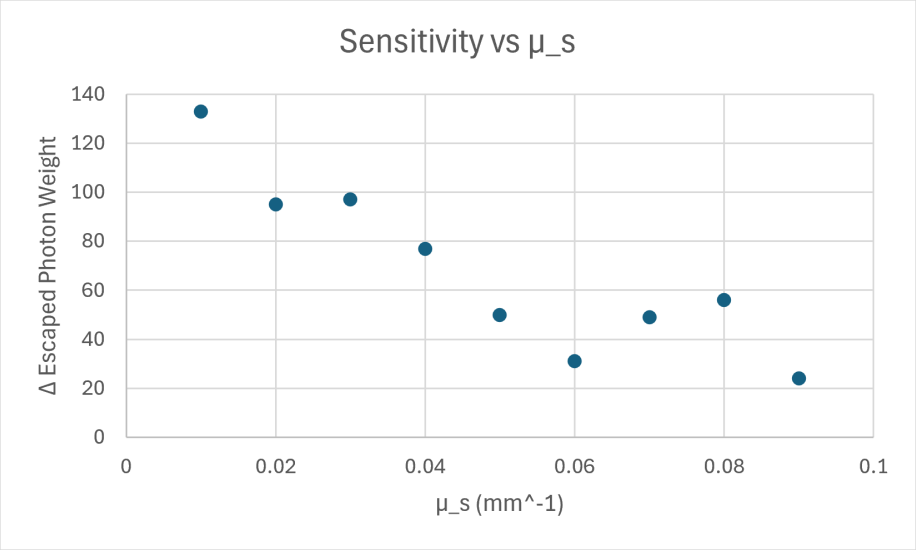


Figure 22: Detector sensitivity. The change in escaping photon weight between consecutive μ_s values from Figure 21, plotted against μ_s. Sensitivity generally decreases as turbidity increases.

4.2.2 Relationship between detector height, z and detector sensitivity

Prior, we established that smallest possible detector to beam axis distance maximizes the radiance as expected. In this section, we show that it also maximizes sensitivity by comparing the average sensitivity at $\mu_s = 0.03, 0.04, 0.05$. These values are chosen because they are expected to be the working range of the device. We also consider the relative sensitivity since the absolute readout values may be amplified or attenuated freely to match some desired value. The average relative sensitivity is calculated as follows:

Given $x*$ is the optimal detector position and the sensitivity is defined as

$$S(x^*, \mu_s \to \mu_s + \Delta\mu_s) = D(x^*, \mu_s + \Delta\mu_s) - D(x^*, \mu_s)$$

The average sensitivity over the working range of 0.03 to 0.05 is

$$\bar{S}(z) = \frac{1}{N} \sum_{\mu_s=0.03}^{0.05} S(x^*, \mu_s \to \mu_s + 0.01)$$

The average amplitude over the same range is given by

$$\bar{A}(z) = \frac{1}{N} \sum_{\mu_s=0.03}^{0.06} D(x^*, \mu_s)$$

Then, the ratio of the amplitudes in comparison to the reference amplitude at $z_{ref} = 65$ is

$$R(z) = \frac{\bar{A}(z_{\text{ref}})}{\bar{A}(z)}$$

Which yields the avarage sensitivity as

$$\bar{S}_{\text{rel}}(z) = \bar{S}(z) \times R(z)$$

Table 3: Average sensitivity around $\mu_s = 0.04$

Detector z pos.	Distance to beam	Avg. sensitivity	Avg. relative sensitivity
65 mm	15 mm	74.6	74.6
75 mm	25 mm	33.6	68.9
85 mm	35 mm	15.0	58.5

Figure 23 shows 2 clusters of distributions. The cluster indicated by the purple ellipse represents distributions taken with $z = 65mm$ and the cluster indicated by the green ellipse represents the distributions taken with $z = 75mm$. The proximity of the points for higher z supports our point qualitatively.

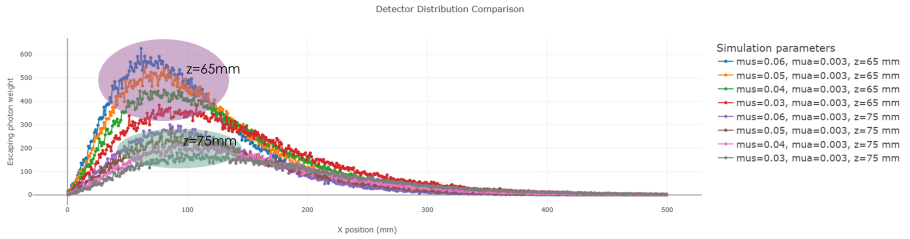


Figure 23: Off-beam detector response profiles for $\mu_s = 0.03$–0.06 mm^{-1} at two detector heights: $z = 65$ mm (purple cluster) and $z = 75$ mm (green cluster). The tighter spacing of the $z = 75$ mm curves indicates lower sensitivity to changes in μ_s.

5 Conclusions

For a classical turbidity meter device given in figure 1, proximity of the positioning of the off the beam detector increases the received radiance as expected. The proximity also increases sensitivity to change in μ_s or turbidity. The following graph tells you the best x position for a given z,and IOPs. Different IOPs and z values may be explored in the following link with an interactive plotter: https://kaanvolkan05.github.io/TurbidWater.

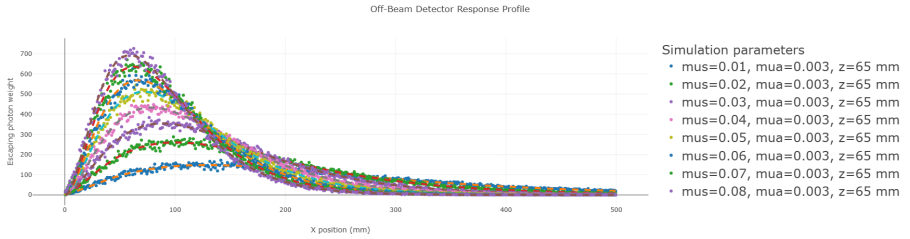


Figure 24: Off-beam detector response profiles for $\mu_s = 0.01$–0.08 mm^{-1} at $z = 65$ mm and $\mu_a = 0.003$ mm^{-1}. Additional IOP and z combinations are available in the interactive plotter.

In generating this database of values, using mcx and a supercomputer containing 8 GPGPUs proved invaluable for itterative exploration of the design space. The details of the device are given in [referance Professor Brorson's website].

6 Validation

Validation tests were employed to ensure the correctness of the simulation and the additional methods. Since a similar example with the same input domain and output range were not found, a number of sanity checks were preformed and compared with theory. In this section, Setup 1, shown in Figure **??** was used unless otherwise specified. Often analysis was preformed by measuring the values on a 2-D cross-section of the system on a circle as shown in figure 25

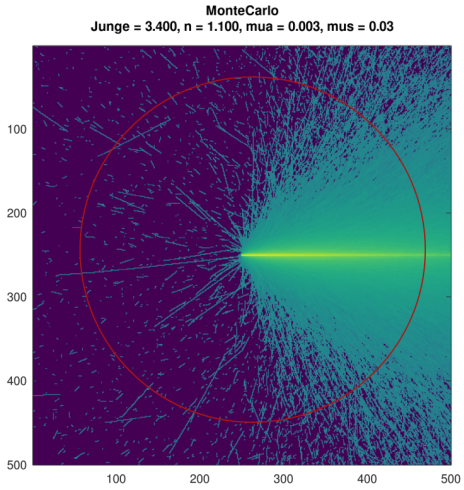


Figure 25: Fluence cross-section through the source plane in Setup 1. The red circle marks an example sampling region for fluency as a function of angle at a fixed radius.

6.1 No Scattering

No scattering, $\mu_s = 0$ must yield a peak intensity on top of the beam line and 0 intensity elsewhere. It must also show intensity reducing along the beam line in accordance to beer-lambert law. Running this simulation with various B_{FF},μ_a and simulation parameters yield the expected results. The figures 26, and 27 below show the results for a given set of parameters.

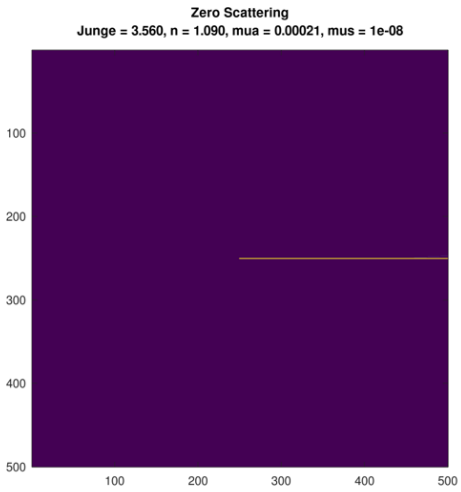


Figure 26: Zero-scattering validation as fluence cross-section in Setup 1 All fluence remains confined to the beam line, as expected.

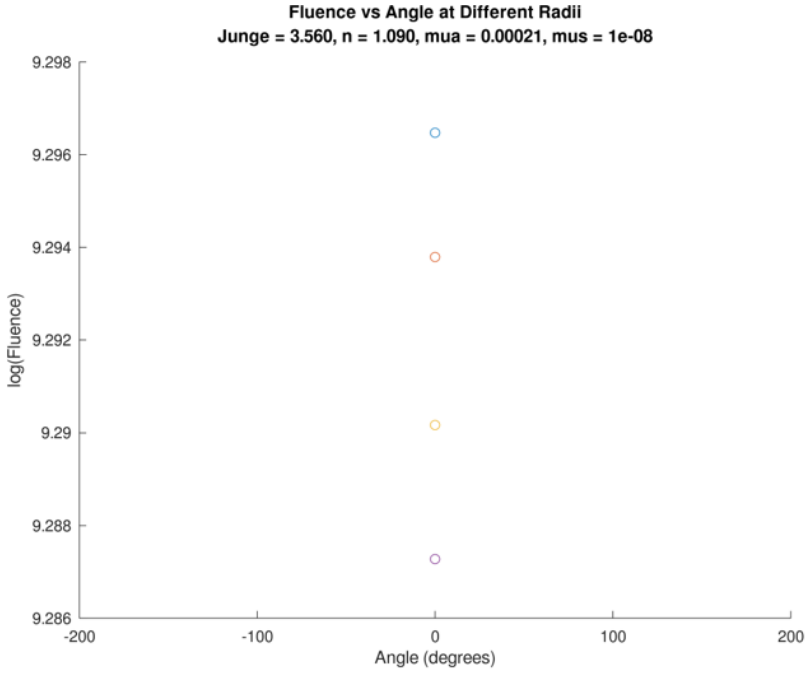


Figure 27: Log fluence vs the sampling angle for the previous figure 26

The agreement between the exponential decay as a function of mu_a is shown when the theory and simulation are compared in the table and graphs below.

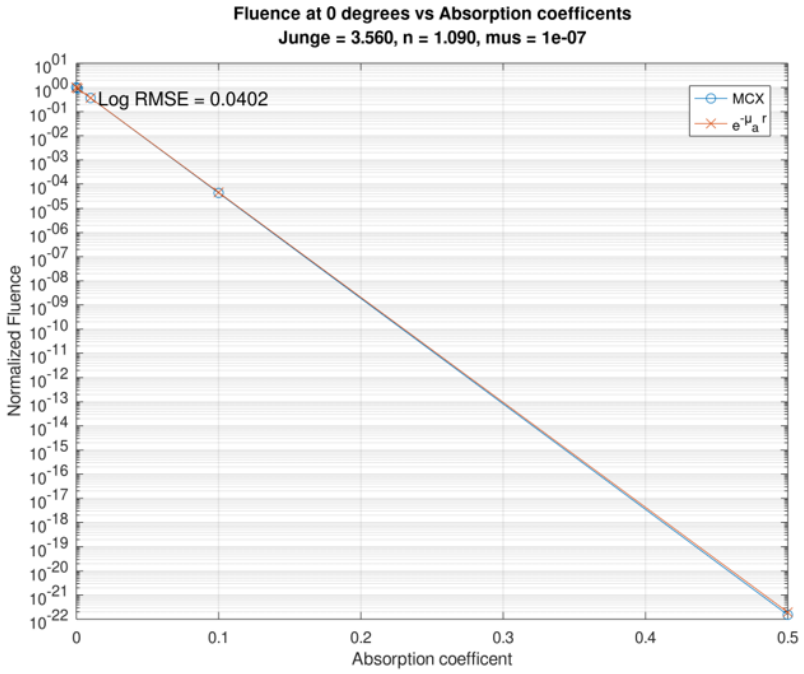


Figure 28: Normalized fluence on the beam line versus absorption coefficient for MCX compared with the Beer–Lambert prediction $e^{-\mu_a r}$, with $\mu_s = 10^{-7}$ mm^{-1}. Log RMSE = 0.0402.

Absorption Coeff	**Normalized Fluence**	**Beer–Lambert**
1×10^{-7}	1.000	1.000
1×10^{-5}	1.007	0.999
1×10^{-4}	0.990	0.990
1×10^{-3}	0.904	0.904
1×10^{-2}	0.366	0.367
0.1	4.32×10^{-5}	4.54×10^{-5}
0.5	~ 0	~ 0

Table 4: Comparison of normalized fluence and Beer–Lambert predictions for different absorption coefficients.

6.2 Under small scattering

6.3 High Scattering

At very high scattering region we expect to see the simulation results resembling an isotropic source despite using a directional source. The figures below qualitatively and quantitatively show the near uniform spread of light around the source.

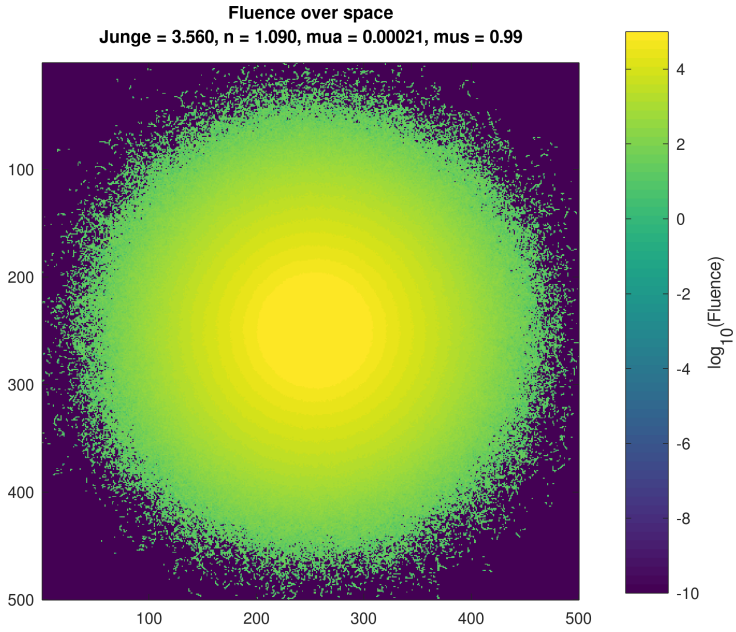


Figure 29: High-scattering validation: $\log_{10}$ fluence cross-section in Setup 1 with $\mu_s = 0.99$ mm^{-1}. Despite the directional pencil-beam source, the light distribution is nearly isotropic.

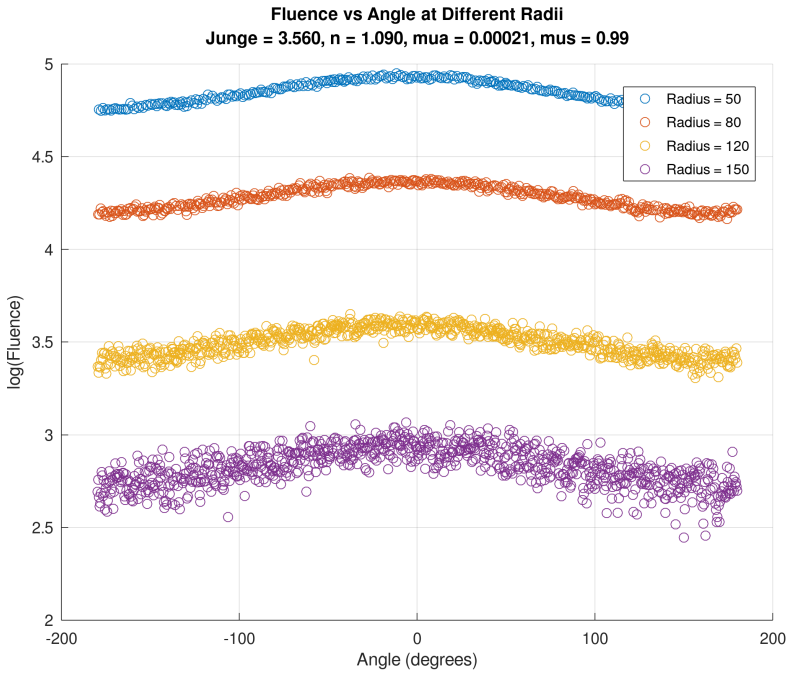


Figure 30: Log fluence vs the sampling angle for the previous figure 29

6.4 Validation via solving the 1D RTE numerically

In order to ensure the Monte Carlo simulation matches the numerically solved problem, we designed and ran a simpler 1D simulation using both MCX and LLLab.

Table 5: Monte Carlo Parameters - Setup 3

Parameter	Value
Number of photons	10^7 (for MC only)
Domain size	200 voxels
Voxel size	1 mm
Source position	0
Source direction	$(1, 0, 0)$
Absorption coefficient, μ_a	0.003
Scattering coefficient, μ_s	0.03
Phase function	Henry Greenstein (g=0.8)
Boundary condition	Cyclic
Time gate	0–5 ns

LLLab solves for the 1D RTE and yields the Total transmitted(downward)

flux and reflected(upward) flux alongside some additional values as a function of distance. The output for the given setup is $R_{disort} = 0.139972$, $T_{disort} = 0.254625$, $R + T = 0.394597$ which means absorbance $= 0.605403$). The same simulation ran on MCX yields the following values, $R_{MCX} = 0.13984 \pm 1.0968e-04$, and $T_{MCX} = 0.25470 \pm 1.3778e-04$ and absorption $= 0.605$. These values are all within expected bounds.

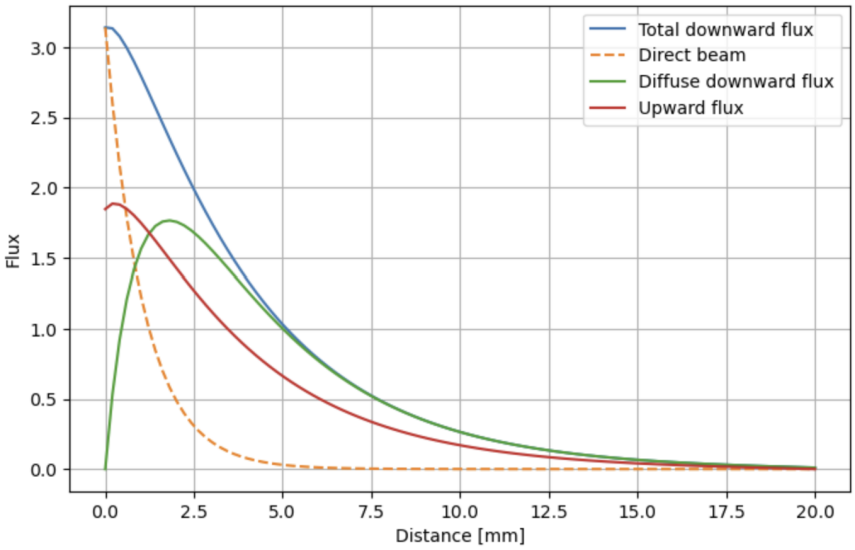


Figure 31: LLLab output graph showing the transmitted(downward) and reflected(upward) flux

7 Referances

References

[1] C. D. Mobley, E. Boss, and C. Roesler, "The Henyey–Greenstein Phase Function," *Ocean Optics Web Book*.

[2] C. D. Mobley, E. Boss, and C. Roesler, "The Fornier–Forand Phase Function," *Ocean Optics Web Book*.

[3] Q. Fang and D. A. Boas, "Monte Carlo simulation of photon migration in 3D turbid media accelerated by graphics processing units," *Optics Express*, vol. 17, no. 22, pp. 20178–20190, 2009.